Road Vehicle Detection Using Fuzzy Logic Rule-based Method

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***Abstract*—Road vehicle detection using very high-resolution remote sensing images has a unique advantage of covering a large area at the same time over all ground-based detectors. But the detection of small vehicle-object in remote sensing imagery is still a challenging task. A scheme was proposed to detect road vehicle objects from airborne color digital orthoimagery based on fuzzy logic rule base. Firstly, a vector-generated road mask was used to constrain detection of vehicles to road region. Secondly, image segmentation algorithm was performed to form image objects in the preprocessing orthoimagery. Finally, based on a set of fuzzy logic rules defined by membership functions, vehicle objects were detected and separated from other objects. A representative set of road segment images was selected from available images to test the proposed scheme. Experimental results indicate that the detection rates of all test road-segments are high with very few false alarms.**

Keywords-remote sensing; vehicle detection; segmentation; fuzzy logic; classification

I. INTRODUCTION

With the availability of sub-metric resolution remotely sensed imagery, the automatic detection, characterization and monitoring of traffic vehicles using airborne or satellite imagery with its unique wide-area coverage, synoptic view has become an emerging field of research [1-3]. Currently, more reliable vehicle detection can only be achieved from the very-high-resolution aerial remote sensing images. Existing approaches for vehicle detection can be categorized based on the underlying type of vehicle modeling [4-7]. There are two basic kinds of vehicle models that have been used: (1) an appearance-based implicit model by clustering appropriate pixel groups into potential vehicles, and (2) an explicit model. The implicit model typically consists of intensity or texture features surrounding each pixel. Detection is performed by checking feature vectors surrounding image pixels. An explicit model usually describes a vehicle as a box or wire-frame representation. Matching the model top-down to the image, or grouping extracted image features bottom-up to create structures similar to the model is performed on the image.

In this paper, we developed a scheme based on fuzzy logic rule base to detect road vehicles from airborne color digital orthoimagery at a ground pixel resolution of 20cm. The basic difference, especially when compared with previously developed pixel-based vehicle detection procedures, is that we don't process and analyze image pixels, but rather image objects that are extracted from image segmentation. In comparison to a single pixel, an image object offers substantially more information. Beyond purely gray-level information, image objects contain a lot of additional attributes which can be used for classification: shape, texture and a whole set of relational/contextual information (semantic information). Fig.1 shows the proposed scheme for road vehicle detection. Firstly, a vector-generated road mask was used to constrain detection of vehicles to road region. Secondly, image segmentation algorithm was performed to form image objects in the preprocessing orthoimagery. Finally, based on a set of fuzzy logic rules defined by membership functions, vehicle objects were detected and separated from other objects by fuzzy classification system.

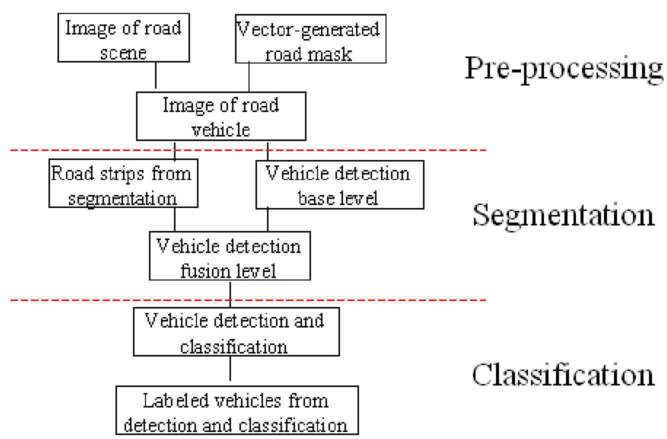

Fig.1 Scheme for road vehicle detection

II. SEGMENTATION

The image segmentation approach used in this study follows the region-merging Multiresolution Segmentation algorithm given in Baatz and Schape [8,9]. All pixels are initialized as single segments that are merged as the algorithm progresses. Segments are merged together when increases in heterogeneity formed by the merger are less than a unitless user-defined threshold of spectral and shape heterogeneity called the scale parameter. As the scale parameter is increased, the size of segments correspondingly increases. Segments are located as potential candidates for merger by identifying the neighbouring segment with the smallest difference in heterogeneity for each individual segment. The contiguous and homogeneous image segments act as image objects. As such, these segments should be meaningful and ideally outline those objects in the image that are to be extracted.

Heterogeneity is calculated according to user-defined weighting of both spectral and shape characteristic metrics. The spectral variance of each input channel in the image is used to measure spectral heterogeneity, and the separate channels themselves may be weighted according to the contribution of each channel to the overall variance of the image. Shape heterogeneity is measured using the ratio of the segment's perimeter length to the perimeter length of a square segment containing the same number of pixels. The influence of spectral vs. shape heterogeneity is determined by weighting these parameters in ratio to each other; the higher the shape criterion the less that spectral heterogeneity influences the segmentation. When the shape criterion is larger than 0 the influence of smoothness vs. compactness of the segments is also determined by weighting of these parameters by ratio.

Because of the multi-scale behaviors of the objects under consideration, small segments can be aggregated to form larger segments by manipulating these criteria; likewise, larger segments can be split into small segments (Fig. 2). This hierarchical network of semantic object information can be implemented by two strategies: the bottom-up and top-down approaches. In the first case, small segments are generated using relatively small scale parameters. Objects generated at higher levels retain the semantic information of the sub-objects. In the top-down approach, large and coarse segments are generated using a relatively large scale parameter. Objects generated at lower levels are determined by the outlines of the super-objects. Both strategies may be employed in the same segmentation routine.

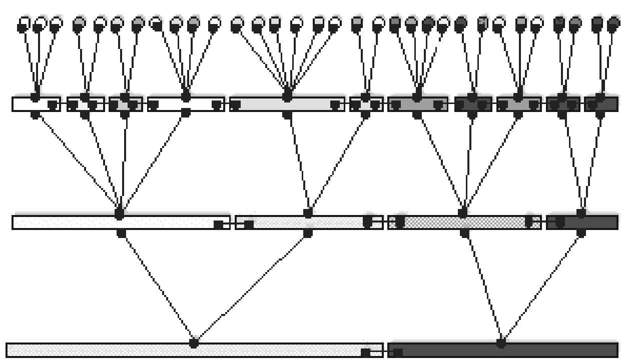

Fig.2 Hierarchical network of objects in segmentation process (Definiens, 2004).

III. FUZZY LOGIC CLASSIFICATION

Once a successful segmentation image is obtained, it is possible to apply a fuzzy logic rule base to discriminate between Vehicle and other Non-Vehicle objects. The goal of the rule base is to represent the uncertain nature of the properties and relationships that characterize Vehicle and Non-Vehicle object classes in membership functions that permit class likelihood to be described. The degree of likelihood to which an object may be assigned membership in a class is evaluated by comparison of fuzzy values for each image object returned by a rule base that quantifiably describes the contribution of specific object features, or sets of features, to an overall likelihood of the object's membership in each class. Class membership for given object features are assigned as scores in which the features' metrics are modeled in membership functions. Membership functions are graphic representations of the proximity of an object's metric for a given feature to a modeled value in a curved or stepped transition between no membership (0) and full membership (1), vice versa, or both. An increasing function is used when membership is assumed to increase with increasing object feature values. Conversely, when membership is assumed to decrease with increasing object feature values, a decreasing function is used. The transitions in membership functions are governed by assigned control points. Fig.3 shows the increasing and decreasing Sigmoidal (*S*-shaped) membership functions, where *a*, *b* and *c* are the control points of the functions. Membership degree μ in class *x* is calculated from an increasing or decreasing sigmoidal function (*S* or 1-*S*), where *a* is the object feature value for attribute *i* at which membership is nil or perfect, *b* is the object feature value at which membership is 0.5, and *c* is the object feature value at which membership is perfect or nil, inverse to *a*.

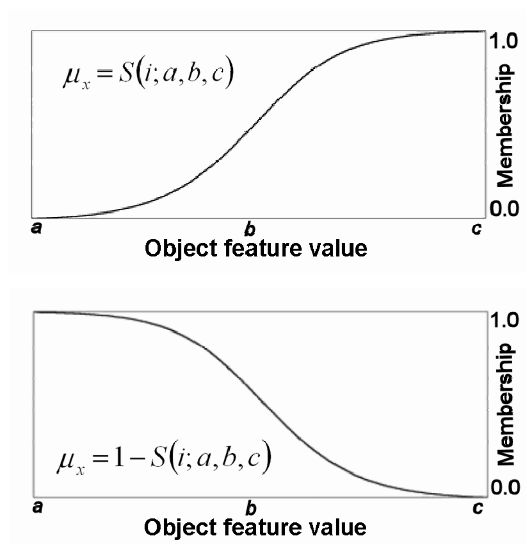

Fig. 3 Sigmoidal (*S*-shaped) membership functions

The high between-class spectral confusion and large within-class spectral variation in the Vehicle and Non-Vehicle object classes makes the supplementary use of spatial, textural and contextual features provided by image segmentation a requirement for classification. Descriptions of the spectral, spatial, textural and contextual features as calculated for objects are given in reference [10]. Because object feature values for spectral, spatial, textural and contextual attributes form the independent variables, or x-axes, of both sigmoidal membership functions and frequency distribution histograms of features, membership in a class itself can be taken to be analogous to frequency for given class attributes. It is therefore possible to assume that distributions of object feature values for independent class attributes that are representative of the class as a whole will be marked by an approximately normal distribution where the accumulation of values within one standard deviation about the mean for each class do not overlap each other.

Since the goal of classification is to identify objects of the Vehicle class, a preliminary rule composed of membership functions for those independent attributes that are representative in the manner described above for the Vehicle class as a whole is used to identify likely Vehicle objects. However, the flat or polymodal distributions of object values for these features in the Non-Vehicle class will result in overestimation of the Vehicle class. A second rule incorporating independent and representative attributes of the Non-Vehicle class as a whole using contextual parameters that describe the relationship between neighbouring class objects is then used to identify likely Non-Vehicle objects from the Vehicle class to reduce the overestimation of the Vehicle class. A third rule using contextual features that describe the relationship between neighboring class objects was then used to further refine the elimination of Non-Vehicle objects from the preliminary Vehicle class. A crisp classification is performed by applying a final rule that delivers to each object a class label assigned by the highest membership value as calculated from application of the first three rules. The iteration of these rules in order forms the fuzzy logic rule base with the spectral, spatial, textural and contextual parameters identified from sample class objects.

In this study, where mean feature values in the modeled class less one standard deviation are greater than mean feature values plus one standard deviation in the other class, the membership function for feature *f* is modeled as an increasing sigmoidal curve:

$$\mu_x = S(i; a, b, c) \tag{1}$$

where *a* is the mean value for the object feature *f* minus one standard deviation in the modeled class, *b* is the mean value minus one-half standard deviation, and *c* is the mean value.

Conversely, where mean feature values in the modeled class plus one standard deviation are less than mean feature values minus one standard deviation in the other class, the membership function for feature *f* is modeled as a decreasing sigmoidal curve:

$$\mu_x = 1 - S(i; a, b, c) \tag{2}$$

where *a* is the mean value for the object feature *f*, *b* is the mean value plus one-half standard deviation, and *c* is the mean value plus one standard deviation.

Membership scores for the modeled class in each rule *R* are combined using an "and" logical operand:

$$\mu_R = \min[\mu_{f_1}, \dots \mu_{f_j}] \tag{3}$$

for f_i for i=1,…j features, or a "or" logical operand:

$$\mu_R = \max[\mu_{f_1}, \dots \mu_{f_j}] \tag{4}$$

and the membership score for the Non-vehicle class is provided as:

$$\mu_{NR} = 1 - \mu_R \tag{5}$$

A crisp classification was performed by applying a final rule that delivered to each object a class label assigned by the highest membership value as calculated from application of the rules. Post-classification merging of adjacent classified objects in the Vehicle and Non-vehicle classes respectively produces a map of vehicle boundaries and locations. Fig.4 shows the image processing stages during vehicle detection. A representative set of 12 road segment images was selected from available images to test the proposed scheme. Fig.5 shows the detection result images of three test road segments. Experimental results indicate that the detection rates of all test road-segments are high with very few false alarms. Table 1 gives the statistical of detection results for the 12 selected road segments (% of true counts).

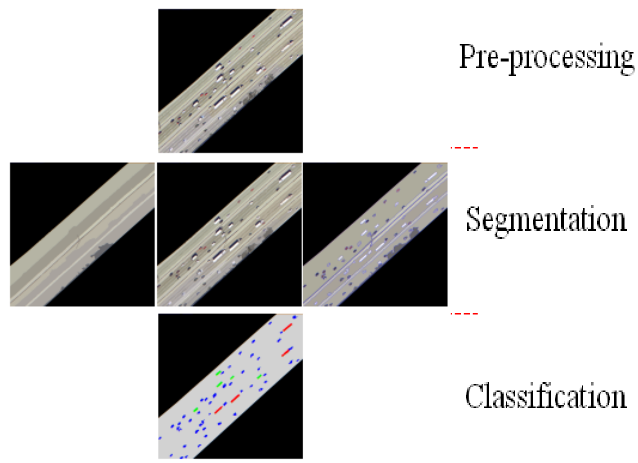

Fig. 4 Image processing stage during vehicle detection

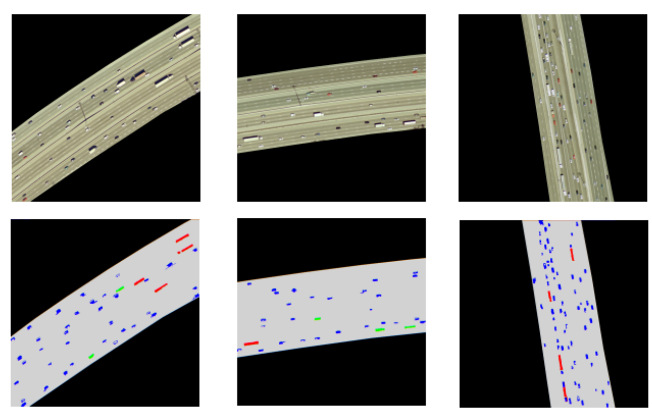

Fig. 5 The detection result images of three test road segments

TABLE I. STATISTICAL OF DETECTION RESULTS FOR THE 12 SELECTED ROAD SEGMENTS (% OF TRUE COUNTS)

Road Segments	Detected vehicles	Omission errors	Commission errors
1	97.6	2.4	0
2	100	1.9	1.9
3	100	0	0
4	102	0	2.3
5	93.7	9.4	3.1
6	106.4	0	6.4
7	97.6	2.4	0
8	100	3.1	3.1
9	97.6	2.4	0
10	93.7	6.2	0
11	105.6	0	5.5
12	102.8	1.4	4.3

IV. CONCLUSION AND DISCUSSION

Experimental results indicate that the proposed method has a good performance. The detection rates of all test road-segments are high with very few false alarms. In our experiments, morphological or post-processing operations are not necessary, which are usually performed on the detected image to sieve out spurious responses and cluster appropriate pixel groups into potential vehicles and finally derives the vehicle-only image. Another advantage is that the proposed method directly detects vehicles on single image; therefore background image is not needed. And further investigation and improvement could include improving automation by incorporating automated road extraction techniques before applying vehicle detection algorithms.

ACKNOWLEDGMENT

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